

LAYER 1: SENSORY & TRANSDUCTION

- Dropped MIPI camera frames (> 100 ms)
- Optical blinding glare / sudden lux drop
- PTP IEEE-1588 clock skew (> 5 ms)
- Depth map sensor dropout / lens mud

LAYER 2: COMPUTE, BUS & MEMORY

- Linux kernel panic / thread crash
- Synthetic DMA memory bus saturation
- malloc allocation stalls & page faults
- Mailbox buffer overrun / lockup

LAYER 3: MODEL & ALGORITHMIC

- Adversarial image perturb / OOD hallucination
- NaN / Inf outputs in diffusion unrolling
- Expired lease execution attempts
- Unbounded jerk / acceleration spikes

LAYER 4: ELECTRICAL & PHYSICAL

- Battery bus voltage sag ($V_{\text{bus}} < V_{\text{min}}$)
- Motor coil thermal overload ($T > 150^{\circ}\text{C}$)
- Sudden external collision load torque
- Encoder phase loss / line break

Camera Drop

Kernel Panic

Hallucination

Voltage Sag

APPLICATION PROCESSOR (Linux MPU · System 2/1.5)

- Runs Vision Transformers, Deliberation & Action Chunking
- Untrusted, stochastic, prone to crashes and latency tails
- Emits candidate trajectory buffer $\mathbf{p}_t$ over shared IPC link

Candidate $\mathbf{p}_t$

REAL-TIME SAFETY ENFORCER (MCU · System 1)

1. **Control Barrier Function Invariant** ($h(x) \geq 0$):
Projects candidate $\mathbf{p}_t$ onto forward invariant safe set $\mathcal{C}$
2. **Dynamic Stopping Clearance Check** ($d_{\text{gap}} > d_{\text{stop}}$):
Vetoes proposals exceeding braking envelope
3. **Zero-Software Hardware Watchdog Monitor:**
Trips Category 1 dynamic stop if MPU heartbeat > 50 ms

Permitted $\mathbf{u}_t$ / E-Stop

PHYSICAL PLANT & ACTUATION ($W_t \rightarrow W_{t+1}$)

- Closed-loop motor current loops · Mechanical brakes · Verified safe stop

PASS CRITERION

100% Containment Rate:

- Zero collisions ($d_{\text{gap}} > 0$)
- Bounded stopping time
- Controlled halt < 50 ms

Zero Uncontained Escapes:

Any single uncontained fault trips a **REFUSE** release verdict.

Hardware Watchdog:

Dedicated non-maskable timer operates independently of Linux OS.